Sorting II - Advanced

Learning Objectives

- Understand the basic principles and algorithms behind merge sort and quicksort.
- Compare and contrast the time complexity of merge sort, quick sort, bucket sort, and cyclic sort, including best-case, worst-case, and average-case scenarios.
- Identify the strengths and weaknesses of each sorting algorithm in terms of stability, adaptability to different data distributions, and ease of implementation.
- Explore potential optimizations and variations of merge sort, quick sort, bucket sort, and cyclic sort, such as parallelization, hybrid algorithms, and memory management techniques.

Lecture Flow

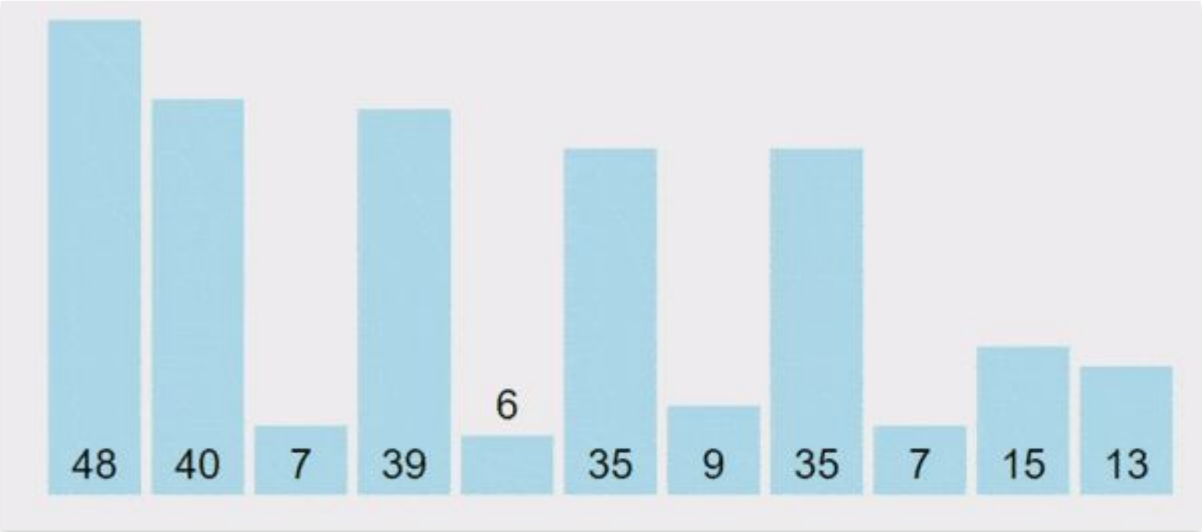
1) Pre-requisites
2) Revision
3) Part I
    - Merge Sort
    - Bucket Sort
4) Part II
    - Quick Sort
    - Cyclic Sort
5) Practice Questions
6) Resources
7) Quote of the Day

Pre-requisites

- Sorting - Basics
- Asymptotic Analysis
- Arrays
- Willingness to learn

Revision

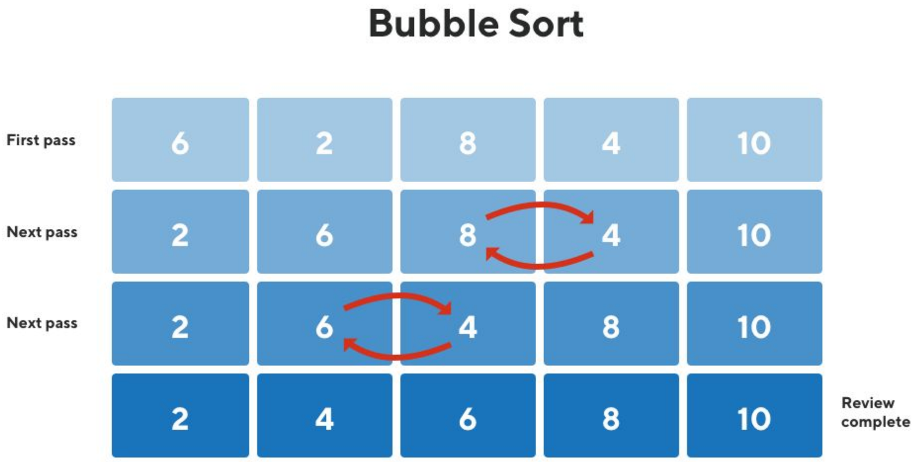
Bubble Sort



Time & Space Complexity

Time complexity ? ____________

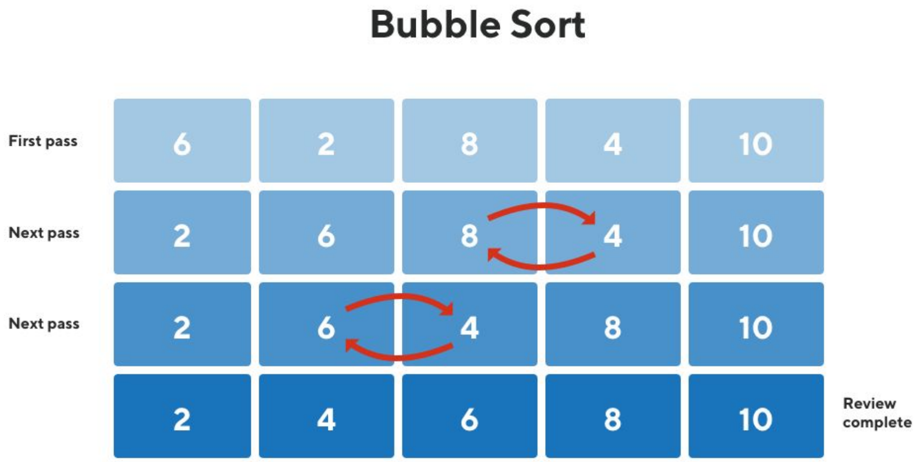
Space complexity ?____________



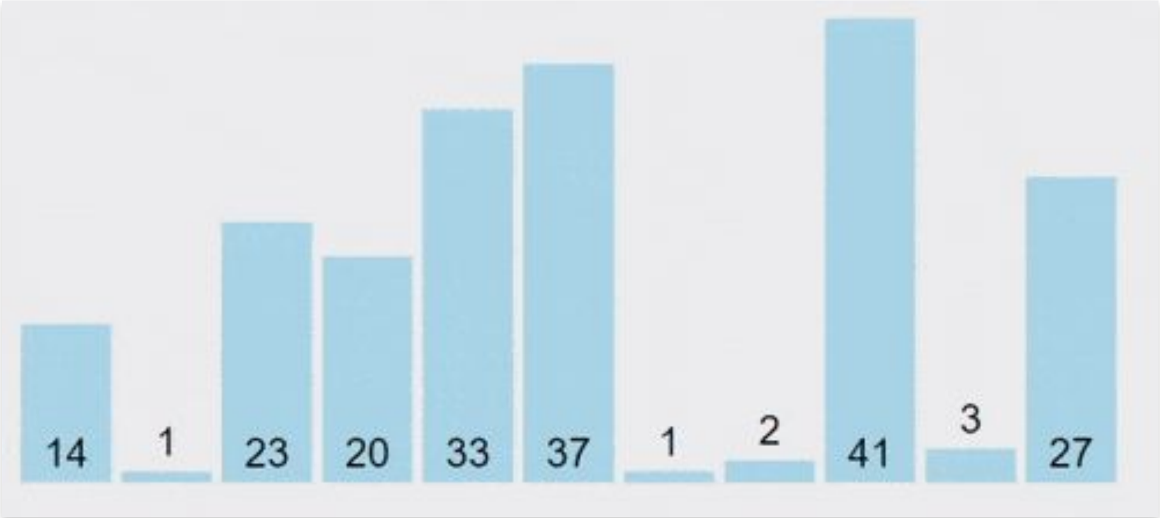
Time & Space Complexity

Time complexity: **O(n^2)**

Space complexity: **O(1)**



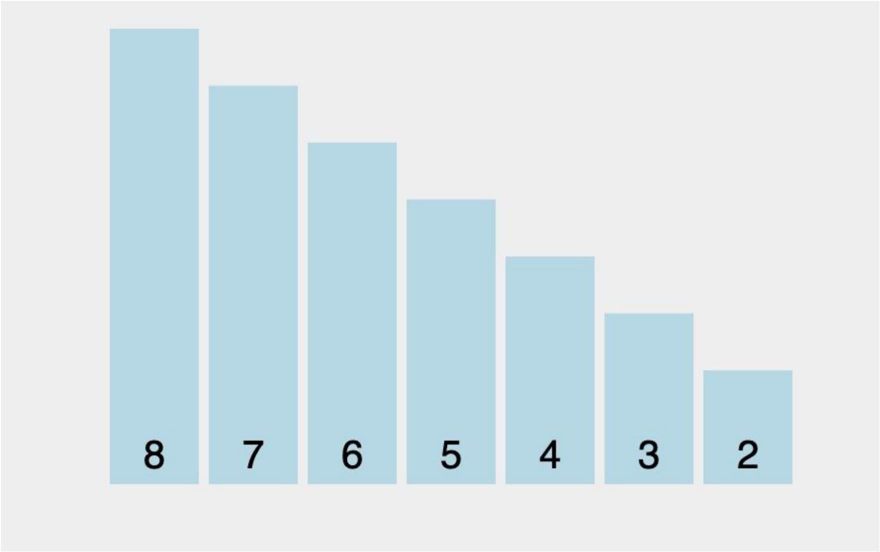
Selection Sort



Time & Space Complexity

Time complexity ? ____________

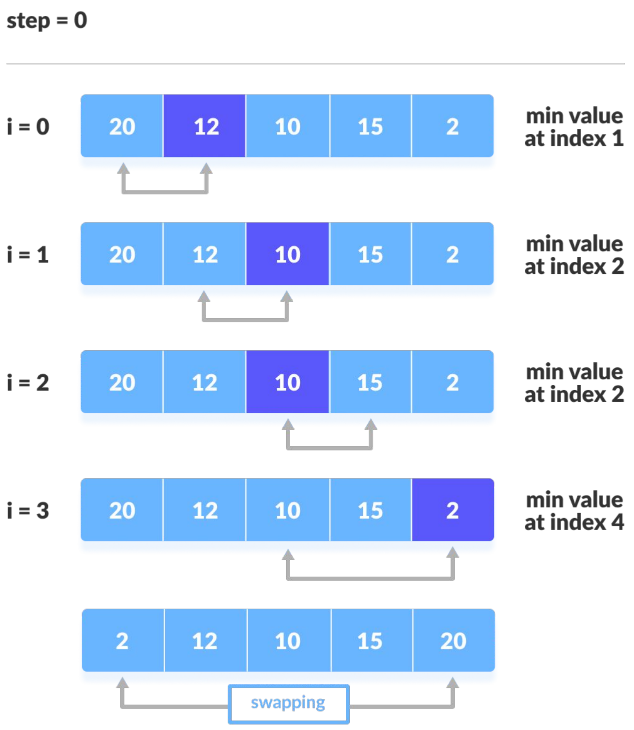
Space complexity ?____________



Time & Space Complexity

Time complexity: **O(n²)**

Space complexity: **O(1)**



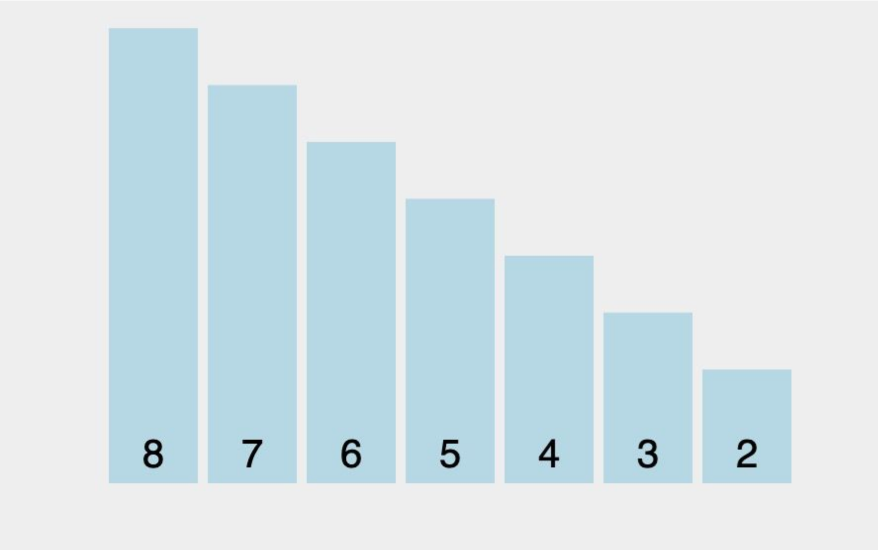
Insertion Sort

Time & Space Complexity

Worst case ? ______________

Best case ? ______________

Average case ? ______________



Time & Space Complexity

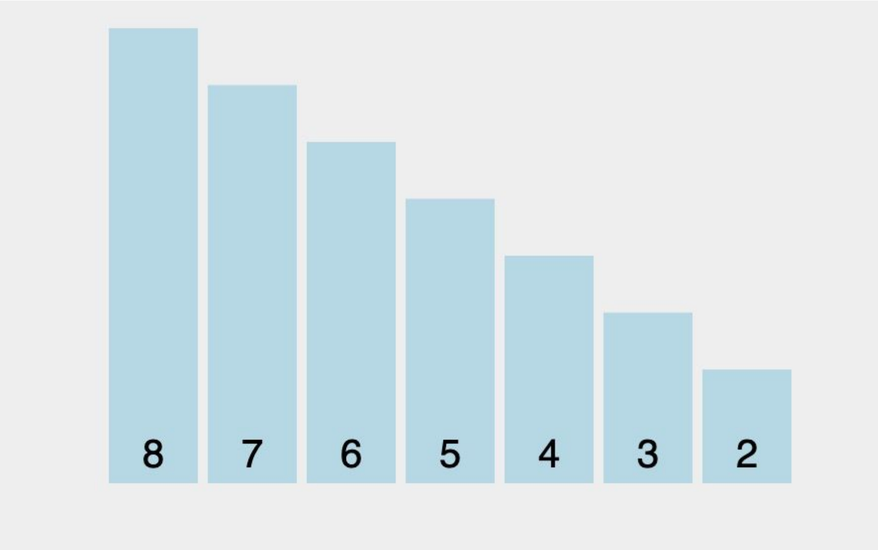
Time complexity: **O(n²)**

Space complexity: **O(1)**

Worst case ___**O(n²)**___

Best case ___**O(n)**___

Average case ___**O(n²)**___



A2SV

Counting Sort

Time complexity

Worst case ? ____________

Best case ? ____________

Average case ? ____________

Time complexity

Time complexity of counting Sort

- Complexity: O(n + k)
  - n = number of elements
  - K = range of values

Worst case - **O(n+k)**

Best case - **O(n+k)**

Average case - **O(n+k)**

📌 **Efficiency Tip: Best when k is not significantly larger than n.**

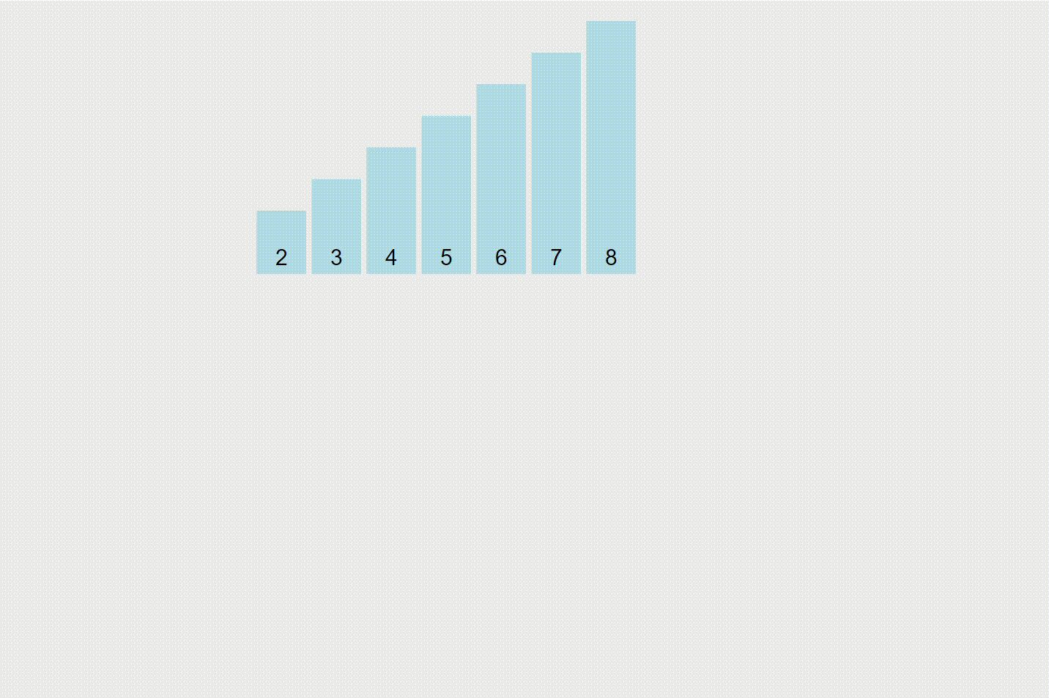
A2SV
Africa To Silicon Valley

Time to get efficient !

Part I

Merge Sort

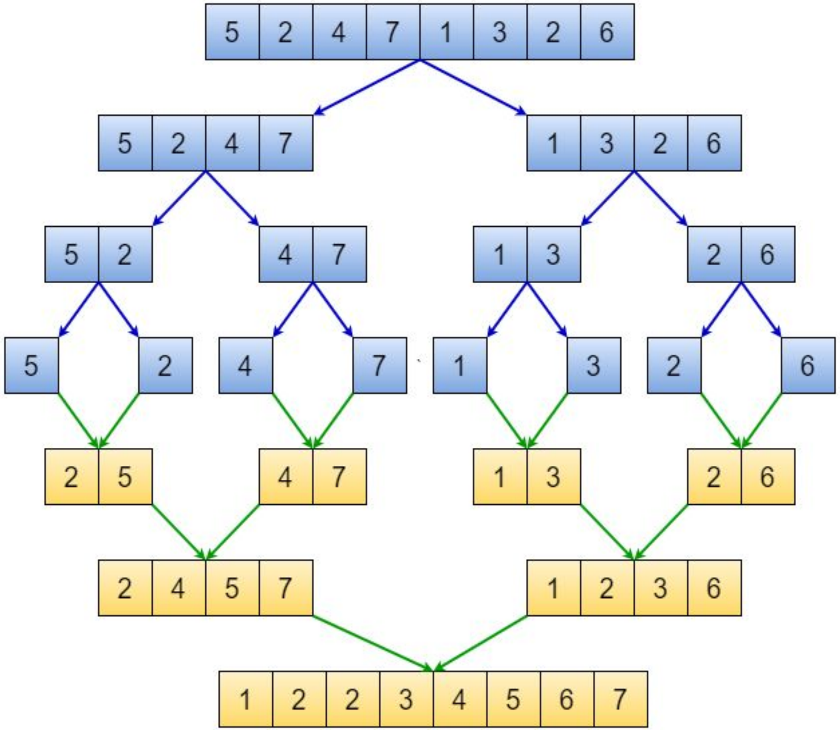
Merge Sort



A sorting algorithm that works by **dividing** an array into smaller subarrays, **sorting each subarray**, and then merging the sorted subarrays **back together** to form the final sorted array.

visualization from: VisuAlgo

Merge Sort



- Divide the array into two halves,
- Sort each half, and then
- Merge the sorted halves back together.

Divide and Conquer

Divide

- Divide the array into two

Conquer

- Sort both halves with merge sort
- Base case?

Combine

- Merge the two sorted halves

Practice

Can you implement the function **merge** ?

Implement Here

Implementation

```python
def merge(left_half, right_half):
    left_index = 0
    right_index = 0
    sorted_subarray = []

    while left_index < len(left_half) and right_index < len(right_half):
        if left_half[left_index] <= right_half[right_index]:
            sorted_subarray.append(left_half[left_index])
            left_index += 1
        else:
            sorted_subarray.append(right_half[right_index])
            right_index += 1

    sorted_subarray.extend(left_half[left_index:])
    sorted_subarray.extend(right_half[right_index:])

    return sorted_subarray
```

Implementation

```
def mergeSort(left, right, arr):
    if left == right:
        return [arr[left]]
    mid = left + (right - left) // 2
    left_half = mergeSort(left, mid, arr)
    right_half = mergeSort(mid + 1, right, arr)

    return merge(left_half, right_half)
```

Q: Is Merge Sort a **Stable** Sorting Algorithm ?

?

Q: What do you think is the time complexity for the aforementioned sorting Algorithm?

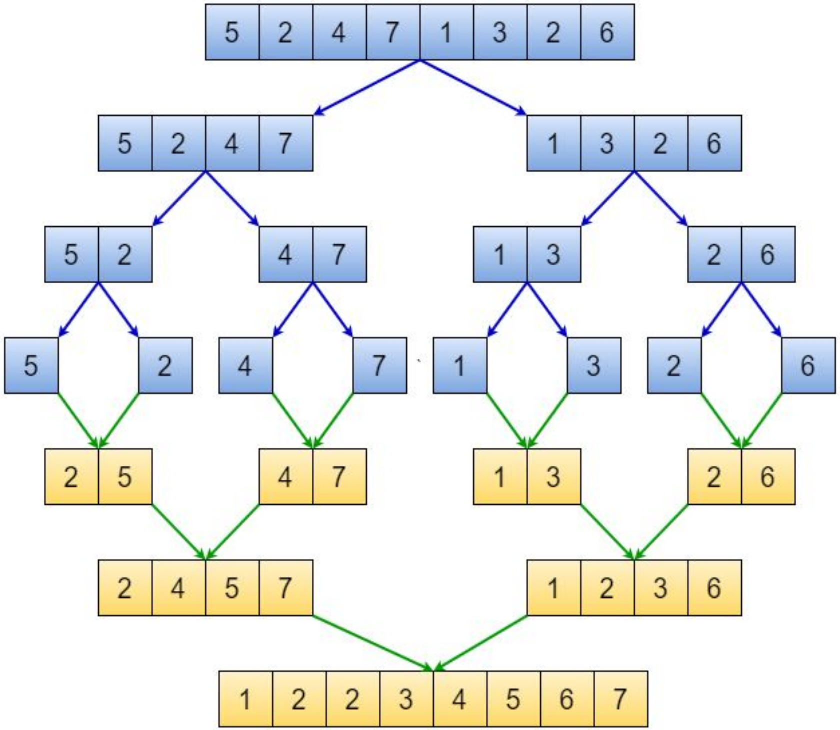
?

Time & Space Complexity

Worst case ________

Best case ________

Average case ________

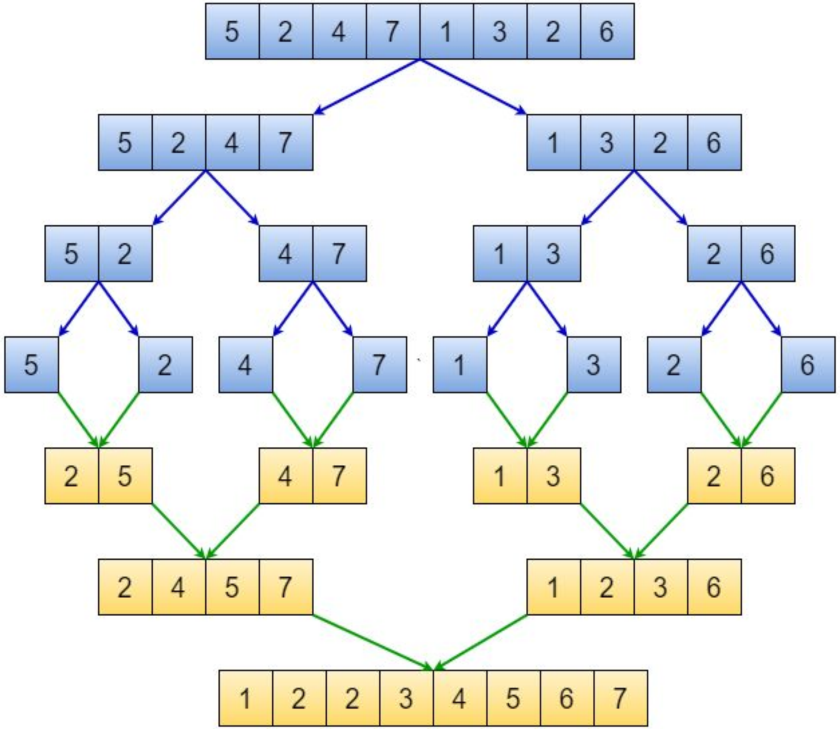


Time & Space Complexity

Time complexity: **O(nlogn)**

Space complexity: **O(n)**

Worst case	**O(nlogn)**
Best case	**O(nlogn)**
Average case	**O(nlogn)**
Stable	**YES**
In-place	**NO**

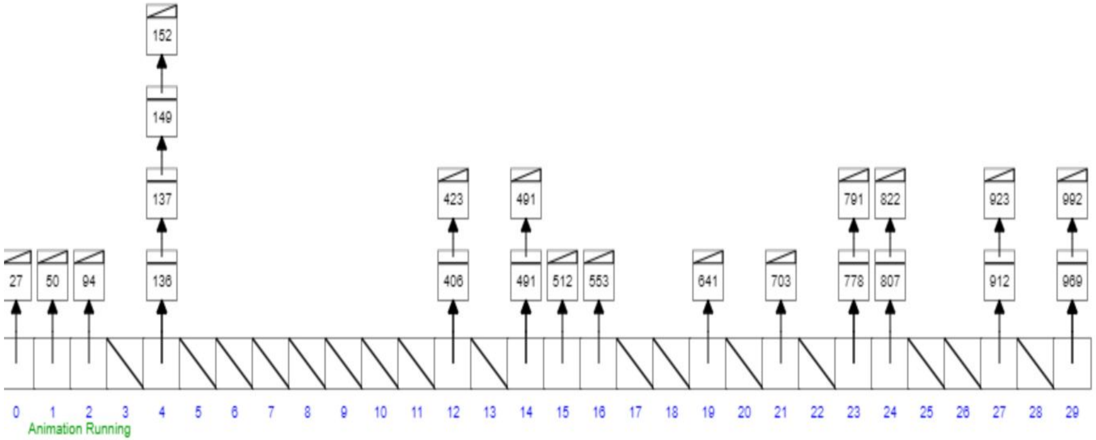


Pair Programming

[Question 1]

Bucket Sort

Bucket Sort



A sorting algorithm that works by **distributing** the elements of an array into a number of **buckets**, and then **sorting each bucket individually**. It is an efficient algorithm for sorting elements that are **evenly distributed across a range of values**.

Bucket Sort

Here's how it works:

1. Determine the **range of values** in the array to be sorted.
2. **Divide** the range into a set of buckets.
3. For each element in the array, determine which bucket it **belongs** to and **insert** it into that bucket.
4. **Sort** each bucket **individually** using another sorting algorithm (usually insertion sort).
5. **Concatenate** the sorted buckets to obtain the final sorted array.

Bucket Sort

Problem:

Sort a large set of floating point numbers which are in **range** from **0.0** to **1.0** and are **uniformly** distributed across the range. How do we sort the numbers efficiently?

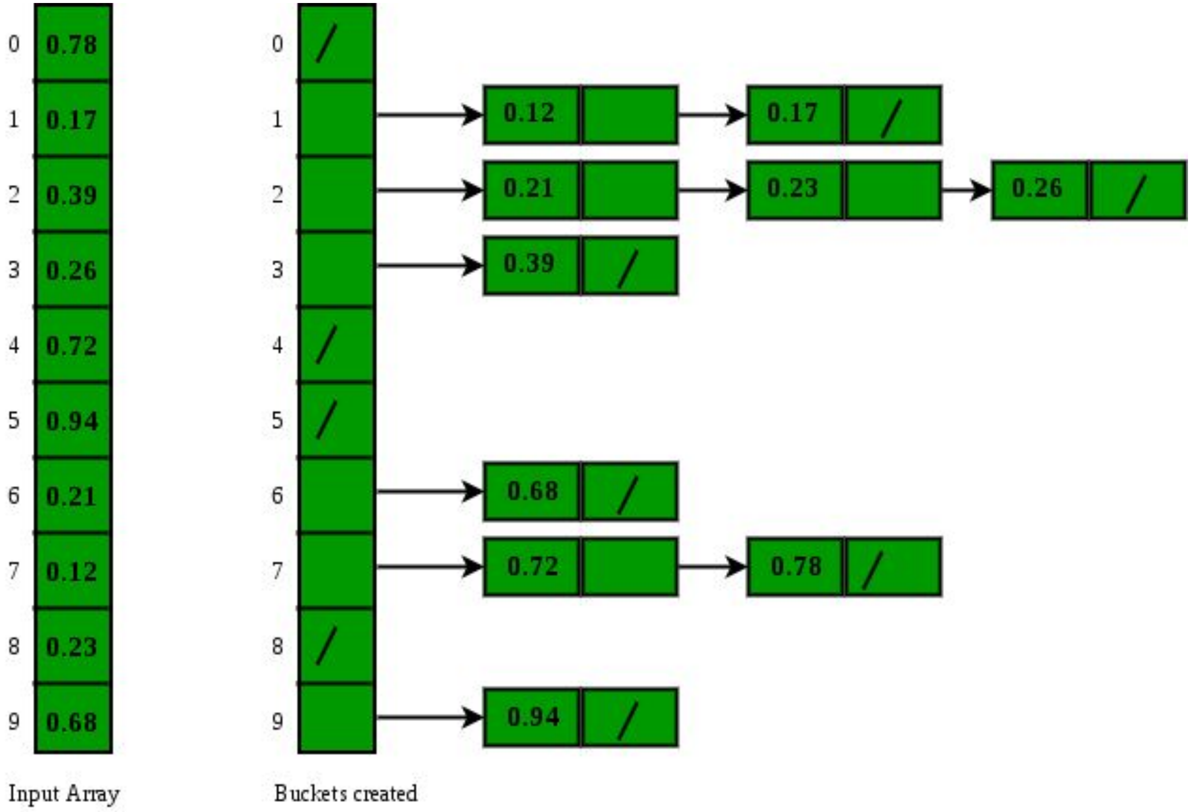
Bucket Sort

Approach:

bucket_sort(arr[], n)

1) Create n empty buckets (Or lists).
2) Do the following for every array element arr[i].
    a) Insert arr[i] into bucket [n*array[i]]
3) Sort individual buckets using insertion sort.
4) Concatenate all sorted buckets.

Bucket Sort


0
1
2
3
4
5
6
7
8
9
0.78
0.17
0.39
0.26
0.72
0.94
0.21
0.12
0.23
0.68
0.12
0.17
0.21
0.23
0.26
0.39
0.68
0.72
0.78
0.94
Input Array
Buckets created

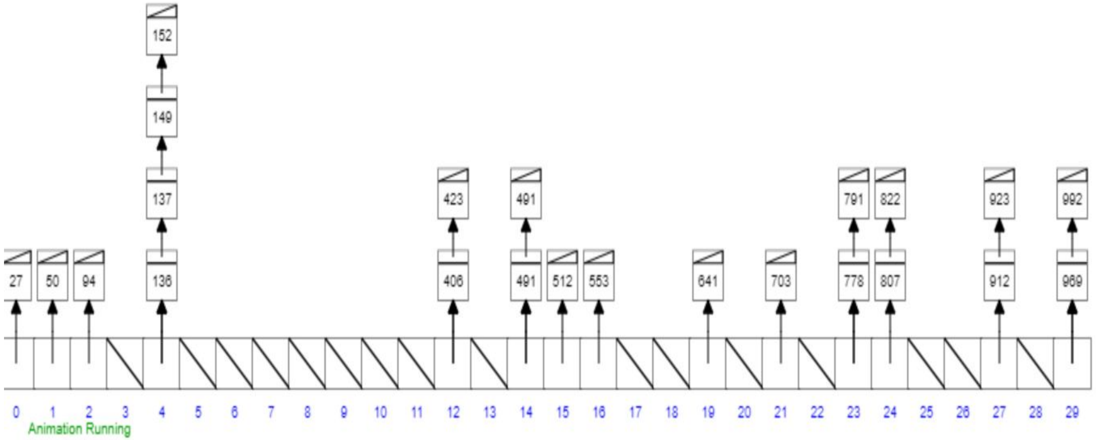
[Visualization Link]

Vanilla Implementation

```python
def bucketsort(arr, n):
    buckets = [[] for _ in range(n + 1)]
    _min= min(arr)
    ans = []
    _range = max(arr) - _min

    if _range == 0:
        return arr

    for num in arr:
        buckets[int(n*(num - _min) // _range)].append(num)

    for elements in buckets:
        ans.extend(insertion_sort(elements))
    return ans
```

Bucket Sort

Bucket Sort



A sorting algorithm that works by **distributing** the elements of an array into a number of **buckets**, and then **sorting each bucket individually**. It is an efficient algorithm for sorting elements that are **evenly distributed across a range of values**.

Bucket Sort

Here's how it works:

1. Determine the **range of values** in the array to be sorted.
2. **Divide** the range into a set of buckets.
3. For each element in the array, determine which bucket it **belongs** to and **insert** it into that bucket.
4. **Sort** each bucket **individually** using another sorting algorithm (usually insertion sort).
5. **Concatenate** the sorted buckets to obtain the final sorted array.

Bucket Sort

Problem:

Sort a large set of floating point numbers which are in **range** from **0.0** to **1.0** and are **uniformly** distributed across the range. How do we sort the numbers efficiently?

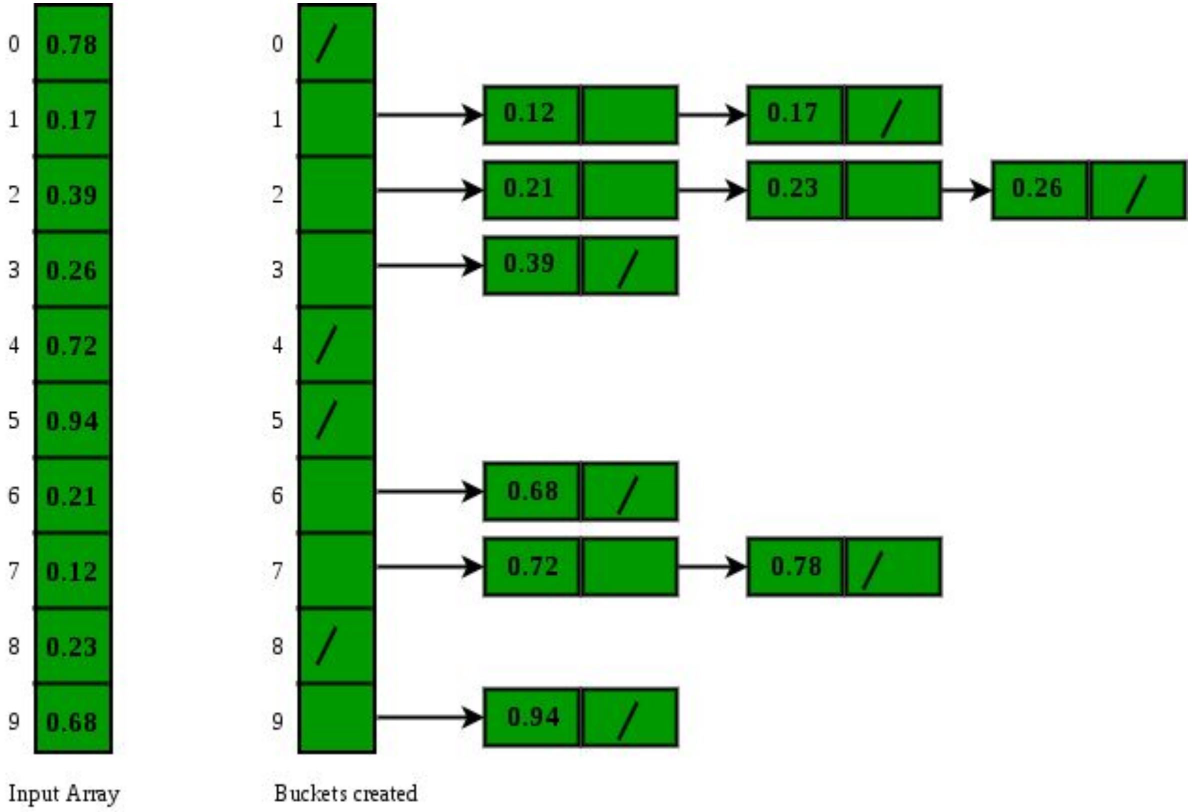
Bucket Sort

Approach:

bucket_sort(arr[], n)

1) Create n empty buckets (Or lists).
2) Do the following for every array element arr[i].
    a) Insert arr[i] into bucket [n*array[i]]
3) Sort individual buckets using insertion sort.
4) Concatenate all sorted buckets.

Bucket Sort



0
1
2
3
4
5
6
7
8
9
0.78
0.17
0.39
0.26
0.72
0.94
0.21
0.12
0.23
0.68
0.12
0.17
0.21
0.23
0.26
0.39
0.68
0.72
0.78
0.94
Input Array
Buckets created

Visualization Link

Can you implement the function **bucket_sort** ?

[**Implement Here**]()

A2SV

Can you implement the function **bucket_sort** ?

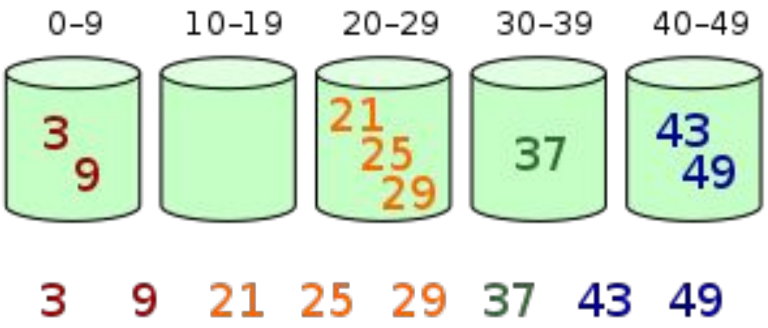
[Implement Here]

Time & Space Complexity

Worst case ? ____________

Best case ? ____________

Average case ? ____________



Time & Space Complexity

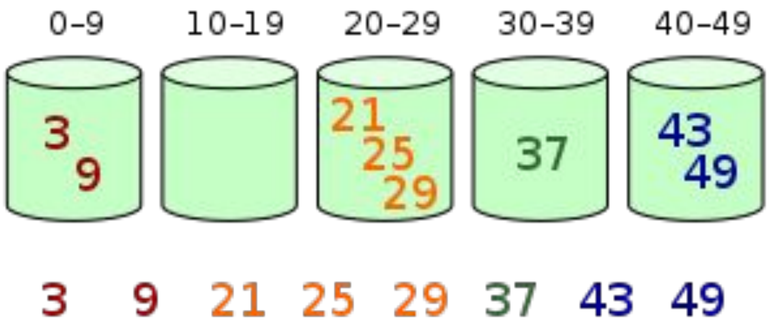
Time complexity: **O(n²)**

Space complexity: **O(n + k)**

Worst case $\underline{\quad}$ **<u>O(n²)</u>**

Best case $\underline{\quad}$ **<u>O(n)</u>**

Average case **<u>O(n + k + (n^2/k))</u>**



Pair Programming

[Question 2]()

Practice Problems

[Sort List]
[Masha and Beautiful Tree]
[Count of Smaller Numbers After Self]
[Number of Pairs Satisfying Inequality]
[Create Sorted Array through Instructions]

Quote of the Day

"The first step in crafting a life you want is to get rid of everything you don't."

- **Joshua Becker**